Geophys. J. R. astr. Soc. (1983) **73**, 129–134

Palaeomagnetism of Upper Cretaceous–Lower Tertiary igneous rocks from Central Argentina

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Received 1982 September 13; in original form 1981 May 11

Summary. From nine Upper Cretaceous–Lower Tertiary (85 ± 5–66 ± 5 Ma) volcanic hills in Central Argentina (33°S, 65°W), 26 hand samples were collected yielding a palaeomagnetic pole at 45°E 70°S ($A_{95} = 12.1°$; $k = 13.6$; $N = 12$) after AC cleaning. Three sites show normal and nine reversed polarity. This pole is close to the pole for the late Cretaceous (69 Ma) Andacolo Series.

1 Introduction

One of the objectives of the Laboratory of Palaeomagnetism of the University of Buenos Aires is the study of Cretaceous rocks from South America. We report the results of a palaeomagnetic study of Upper Cretaceous–Lower Tertiary igneous rocks from Central Argentina.

The geodynamical and geological implications of the South American Jurassic–Cretaceous palaeomagnetic poles are discussed in the following paper.

2 Geology

In the central part of Argentina (about 33°S, 65°W; Fig. 1) some volcanic hills occur in an area of basament rocks. The petrology of the Cerritos Negro (I–V) (32.5°S 65.0°W) has been worked out by Solá (1979). The composition of the rocks of these hills varies from basalt to nepheline basanite. Rocks formed in at least two igneous phases were found in some of these hills. Whole rock K/Ar age determinations carried out on samples from the Cerritos Negro II and V (Table 1) suggest a late Cretaceous age (Solá 1979).

The petrology of the volcanic hills situated to the east of the Mercedes city (33.7°S 65.0°W) has been worked out in detail by López (1979). The rocks from Cerros La Leoncita, La Madera, La Garrapata and La Piedra are mainly volcanic nephelinites. A late Cretaceous–early Tertiary age was suggested for these hills mainly on the basis of whole rock K/Ar determinations (López 1979; Table 1).

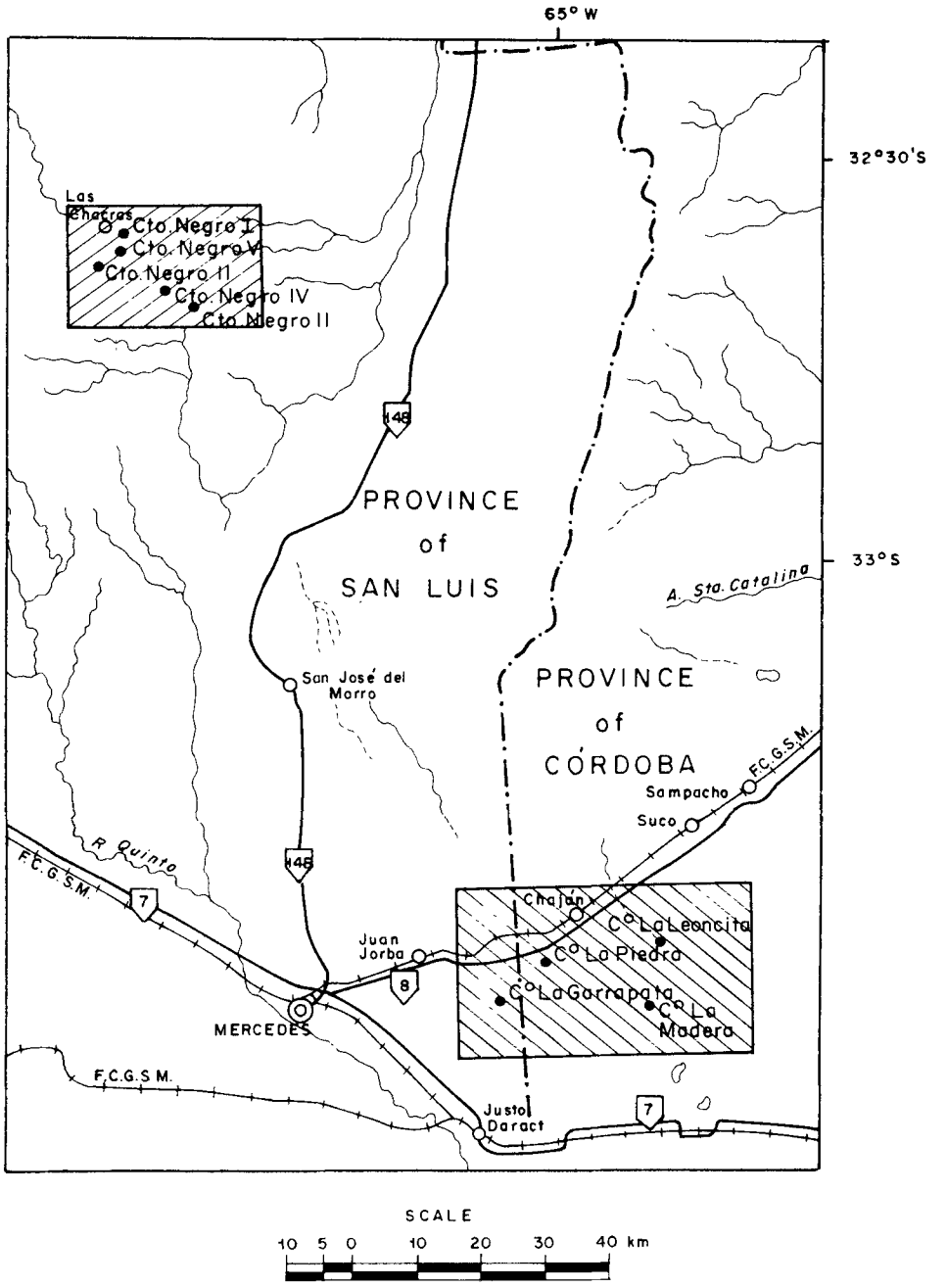


Figure 1. Map showing the location of the Upper Cretaceous–Lower Tertiary volcanic hills (sampling sites) in the central part of Argentina: -+++-, railway; —— main road.

Table 1. Palaeomagnetic and K/Ar data for Upper Cretaceous–Lower Tertiary igneous rocks from Central Argentina.

Zone	Hill	Site or sample	K/Ar age (Ma)	Stable remanent magnetization N	D (°)	I (°)	k	α_{95} (°)	Polarity	Virtual geomagnetic pole Long. (°E)	Lat. (°S)
Las Chacras	Cto. Negro I	77b, 79	–	4	351	−33	51	13	N	84	73
(Province of San Luis)		78	–	2	35	−52	72	30	O_N	215	61
32° 30′S 65°W	Cto. Negro II	100, 103	85 ± 5	4	340	−10	26	18	N	77	57
		101	–	2	170	40	1399	7	R	70	77
	Cto. Negro III	105	–	2	330	−19	1444	7	N	58	54
		108, 110	–	3	40	15	21	28	O_N	166	35
	Cto. Negro IV	116	–	2	320	−36	75	29	N	34	52
		117	–	2	358	−60	105	25	N	307	81
	Cto. Negro V	90	–	2	81	−53	1451	7	O_N	231	24
		95	83 ± 5	2	356	−18	140	21	N	105	66
		96	–	2	32	12	81	28	O_N	159	41
Villa Mercedes–Chajdn	Co. La Leoncita	15, 16, 18	66 ± 5	6	342	−28	34	11	N	70	65
(Provinces of San Luis	Co. La Madera	36.39	–	4	351	−66	10	31	N	317	74
and Córdoba)		40	–	2	149	22	41	40	R	56	54
33°40′S 65°W	Co. La Garrapata	77a	–	2	171	68	418	12	R	313	71
		123, 126	75 ± 7	4	299	−55	11	29	N	4	40
	Co. La Piedra	118	–	2	249	17	243	16	O_R	201	22
		121	–	2	74	−17	554	11	O_N	203	18
									Mean pole position	45	70

($N = 12, A_{95} = 12.1°, k = 13.6$)

N is the number of specimens submitted to AF treatment; k the precision parameter, and α_{95} the radius of the 95 per cent confidence circle (Fisher 1953). D and I are the declination and inclination, respectively.

3 The palaeomagnetic study

3.1 SAMPLING

Fourteen oriented hand samples were collected from five volcanic hills situated to the east and south-east of Las Chacras and 12 samples were collected from four volcanic hills immediately to the east of the City of Mercedes (Fig. 1 and Table 1). Samples were oriented using both sun and magnetic compasses.

3.2 LABORATORY METHODS

Fifty-two specimens (cylinders 2.5 cm in diameter and 1 cm height) were cut. A spinner magnetometer was used to measure the remanent magnetization of these specimens. Alternating field demagnetization (AF) was used to isolate the stable remanence of one pilot specimen of each sample (Valencio, Vilas & Mendía 1977) and the other specimens from each sample were demagnetized in the optimum alternating field deduced for the pilot specimen. The secondary component of remanence, present in most of the samples, was successfully erased by the optimum demagnetizing field which varied between 150 and 200 Oe, at which more than 15 per cent of the NRM was retained (Fig. 2).

Fisher's (1953) method was used to compute the mean directions of stable remanent magnetization for the sites given unit weight to each specimen, and virtual geomagnetic poles (VGPs) were calculated (Table 1). A mean pole was computed and VGPs situated more

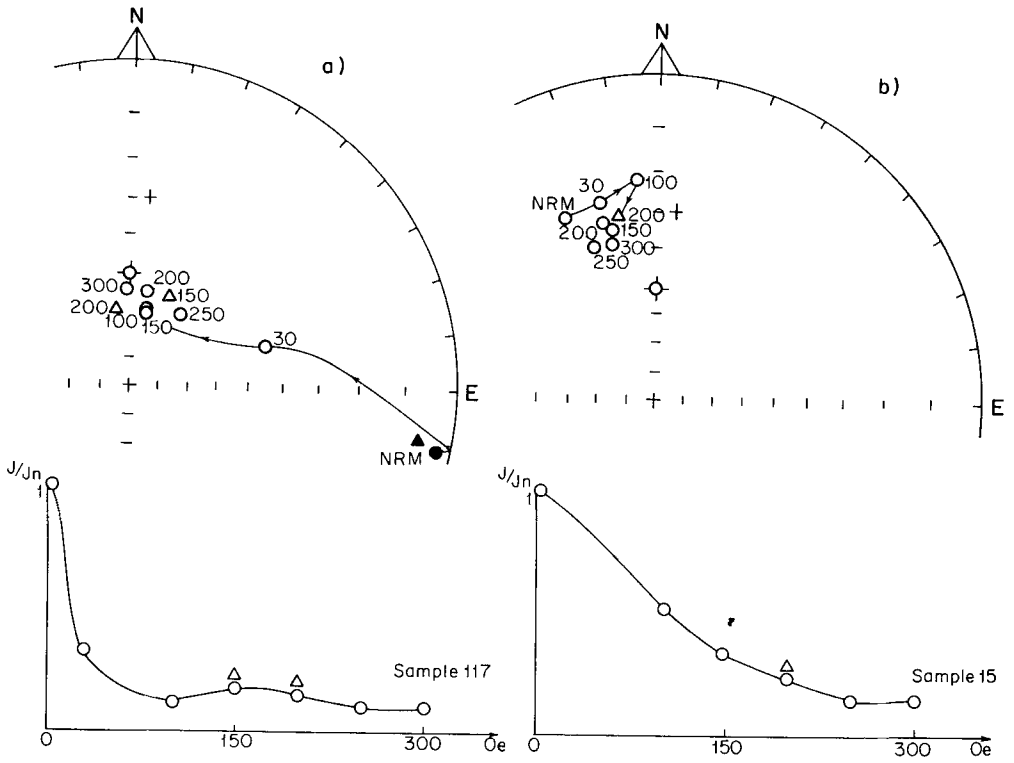


Figure 2. Changes in direction and intensity of magnetization by progressive AF cleaning of the pilot specimen (○) and one other specimen (△) from samples from Cerrito Negro IV (a) and Cerro La Leoncita (b). Solid symbols indicate downward dipping directions. The directions of the present geomagnetic field and the axial dipole are shown by + and ⊹ respectively.

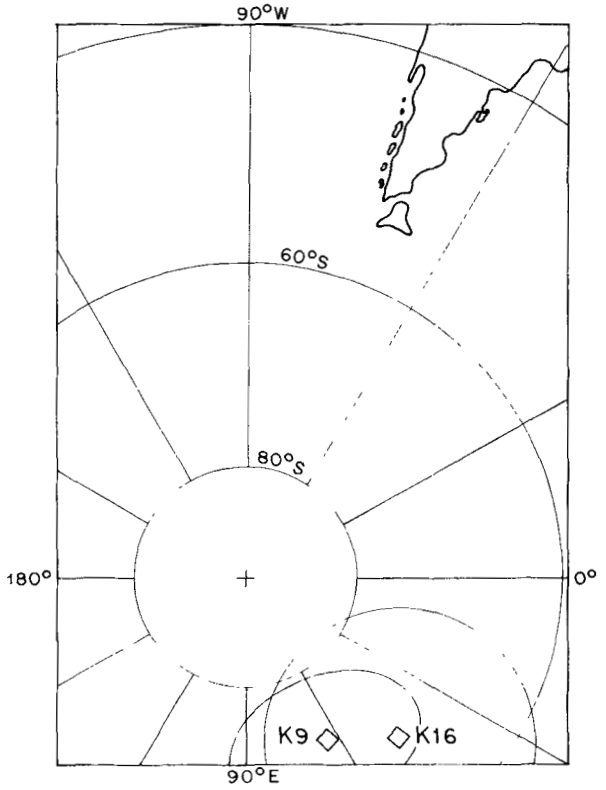


Figure 3. Late Cretaceous–early Tertiary palaeomagnetic poles for South America. K_{16} is the palaeomagnetic pole for the volcanic hills of Central Argentina presented in this paper. K_9 is the PP for the Andacollo Series (Vilas & Valencio 1978).

than 40° from the mean position were rejected (classified as oblique (0) in Table 1); a new mean pole was then calculated. The population of VGPs all within 40° of the mean position, yields a palaeomagnetic pole at 45°E 70°S ($N = 12$, $k = 13.6$, $A_{95} = 12.1°$, $R = 11.193$; SAK_{16}; Fig. 3). Nine sites yielded normal (N) and three reversed (R) mean cleaned remanence direction. Details about the geological significance of the palaeomagnetic data are given in Valencio *et al.* (1980). The pole position given above supersedes those presented in that paper.

4 Discussion

Fig. 3 shows that the position of the palaeomagnetic pole (PP) for the volcanic hills from San Luis and Córdoba Provinces is close to the position of the late Cretaceous PP for the Andacollo Series (SAK_9: 74°S 65°E, $d\psi = 6°$ $d\chi = 9°$; 69 Ma; Vilas & Valencio 1978). This is consistent with the late Cretaceous–early Tertiary age suggested for those igneous rocks by the radiometric study.

Acknowledgments

The authors wish to thank the Universidad de Buenos Aires and the Consejo Nacional de Investigaciones Científicas y Técnicas for their support which enabled the work described to be carried out.

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